

Pharmacology of Antihypertensive Agents: Video series

- **Calcium Channel Blockers**
- **Beta Blockers**

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**I am available to groups and individuals for
pharmacology help and discussions by appointment.**

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Following this session, students will be able to:

1. Interpret the pharmacokinetic properties of the antihypertensive agents in the context of their clinical significance.
2. Relate the mechanisms of action of the calcium channel blockers and beta blockers to their cardiovascular effects by giving examples of therapeutic applications.
3. Predict the adverse effects, drug interactions, and contraindications when given the mechanism of action and/or pharmacokinetic properties of the individual drug classes.
4. Select the appropriate antihypertensive agent and monitoring parameters for the individual patient based on the drug's pharmacologic effects when presented with a clinical case vignette.

Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions and Clinical Vignettes

Highly relevant optional materials:

- Dr. Goldstein's Word handout | Video Lecture | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them.*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions (not graded): Simple Recall and Case Vignettes*

Links to resources

Scholar Rx Brick (Required)

Antihypertensive drugs <https://exchange.scholarrx.com/brick/antihypertensive-drugs>

Renin-Angiotensin-Aldosterone System Inhibitors <https://exchange.scholarrx.com/brick/renin-angiotensin-aldosterone-system-inhibitors>

Helpful bricks on blood pressure physiology (Recommended but not required):

Autonomic regulation of the cardiovascular system <https://exchange.scholarrx.com/brick/autonomic-regulation-of-the-cardiovascular-system>

Hypertension: Foundations and Frameworks <https://exchange.scholarrx.com/brick/hypertension-foundations-and-frameworks>

Primary (Essential) Hypertension <https://exchange.scholarrx.com/brick/primary-essential-hypertension>

Baroreceptor regulation of blood pressure <https://exchange.scholarrx.com/brick/baroreceptor-regulation-of-blood-pressure>

Humoral regulation of blood pressure <https://exchange.scholarrx.com/brick/humoral-regulation-of-blood-pressure>

Textbooks/Exam review books you might find helpful:

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024

Chapter 11: Antihypertensive Agents

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281748109>

Chapter 17: Vasoactive Peptides > Angiotensin

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281749431>

LWW Health Libraries Premium Basic Sciences Lippincott Illustrated Reviews: Pharmacology, 8e, 2022; Chapter 8: Antihypertensives

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253325981&bookid=3222>

Access Medicine Katzung's Pharmacology: Examination and Board Review, 14e, 2024; Chapter 11: Drugs Used in Hypertension

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285590661>

- For tips on learning skills based on the science of learning and for “What you need to know”, please see RAS Inhibitors lecture, slides 6-9.

L-TYPE CALCIUM CHANNEL BLOCKERS (CCBs)

Non-DHP	DHP, Oral	DHP, I.V.
* Verapamil	* Nifedipine	* Nicardipine
* Diltiazem	* Amlodipine Several others	* Clevidipine

CCBs: Pharmacokinetics (Class Properties)

Good oral absorption but reduced bioavailability due to hepatic first-pass effect

High plasma protein binding; many are P-glycoprotein substrates

CYP-mediated metabolism, esp. CYP3A4

$t_{1/2}$ vary between 1.3 – 64 hours (drug-specific property)

- ***Extended release forms reduce dosing frequency of short-acting agents***

Selected CCBs: Pharmacokinetic Properties

DRUG	BIOAVAILABILITY (immediate release)	ONSET OF ACTION	PROTEIN BINDING	METABOLISM	PLASMA t _{1/2} (normal organ function)
<i>Non-Dihydropyridines</i>					
Verapamil	20-35%	Oral 1-2 h I.V. 1-5 min	~90%	CYP3A4 (multiple others)	4.5-12h
Diltiazem	40%	Oral 30-60 min I.V. 3 min	~75%	CYP3A4 (and others)	3-4.5 h
<i>Dihydropyridines</i>					
Nifedipine (oral)	40-77%	~20 min	>90%	CYP3A4	2-5 h
Amlodipine (oral)	64-90%	24-48 h	>90%	CYP3A4	30-50 h
Nicardipine (I.V.) Clevidipine (I.V.)	n/a	minutes	>95%	CYP3A4,2C8, 2D6	α 3 min, β 45 min, terminal 14h only after long infusion
Nimodipine	13%	Oral ONLY	~95%	CYP3A4	1-2 h

Some Pharmacokinetic Concepts Related to CCBs

Extended-release forms reduce the number of daily doses needed to maintain therapeutic drug levels.

Example:

- Nifedipine immediate release 3x daily (every 8 hours)
- Extended release 1x daily

Nifedipine extended-release appears to cause less reflex tachycardia than the immediate release formulation

Amlodipine has slow absorption, minimal peaks and troughs, and prolonged effect.

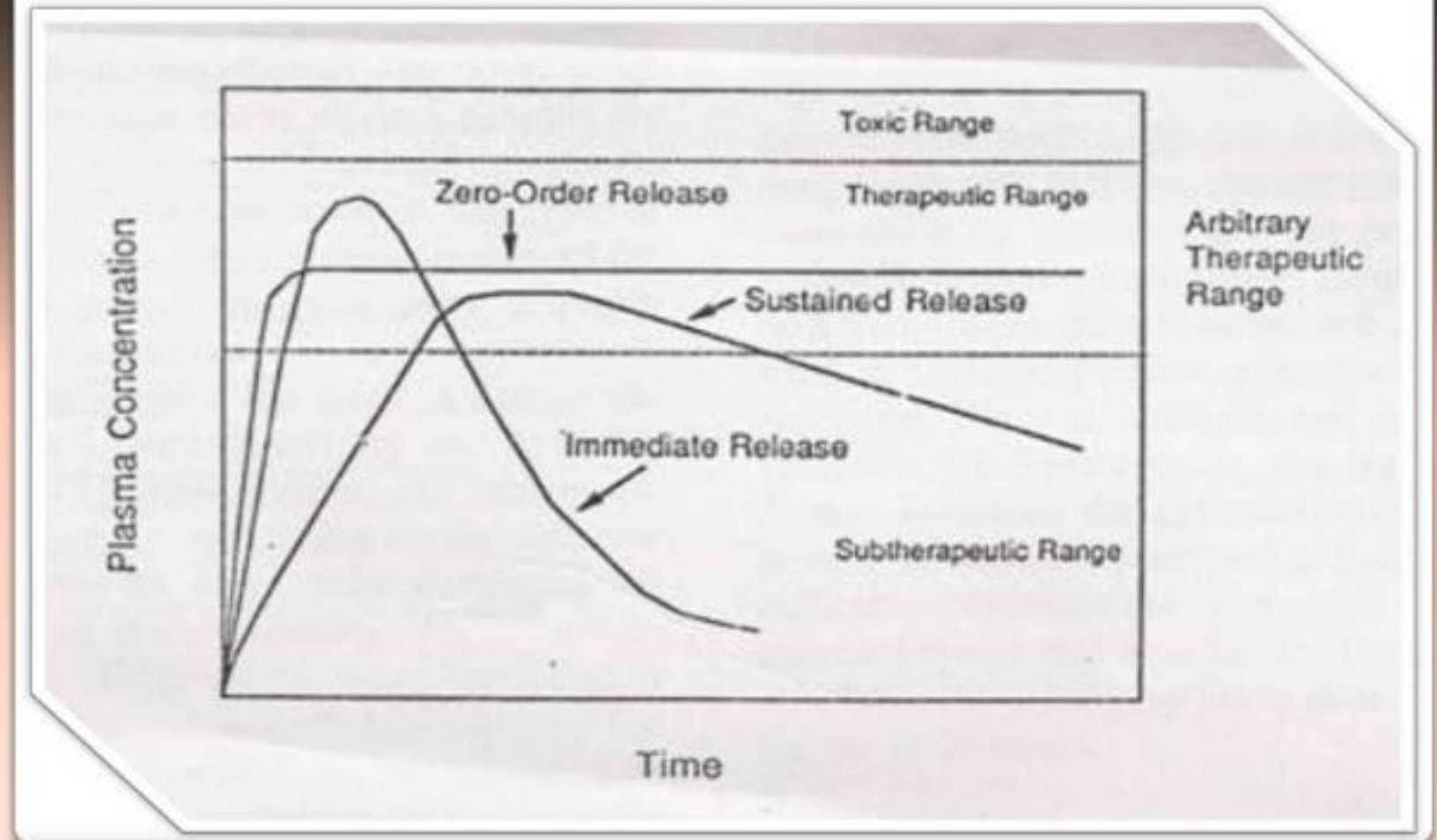
It causes less reflex tachycardia, possible due to lower peaks and long $t_{1/2}$.

Immediate release DHP CCBs with short half-lives are not recommended.

Within each dosage interval, they can cause:

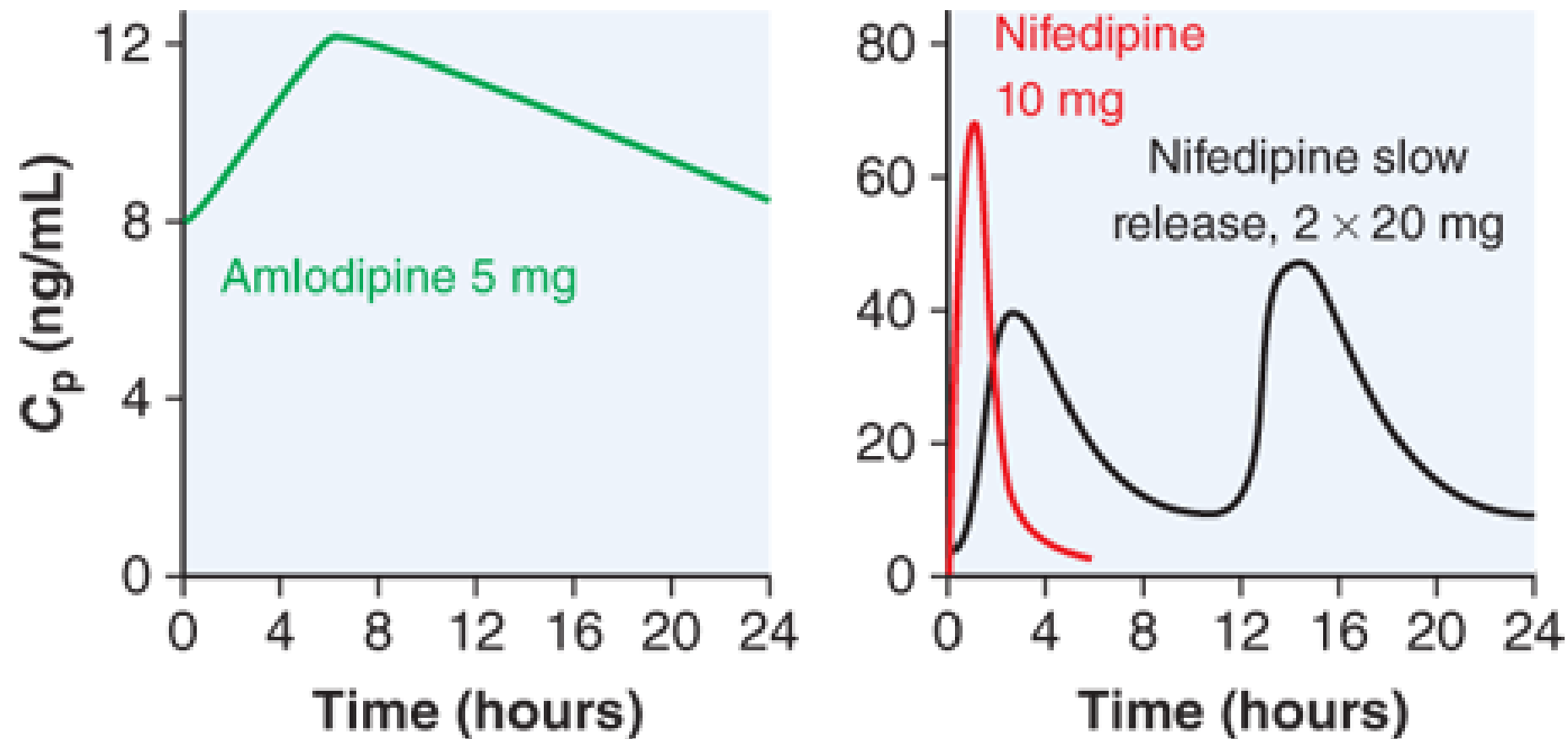
- 1) Rapid drop in blood pressure,
- 2) Oscillations in blood pressure control, and
- 3) Concurrent surges in sympathetic reflex activity.

Plasma concentration v/s time curve



<http://www.slideshare.net/prashantmane01/sustained-release-drug-delivery-system>

Zero-order release = constant rate of drug release



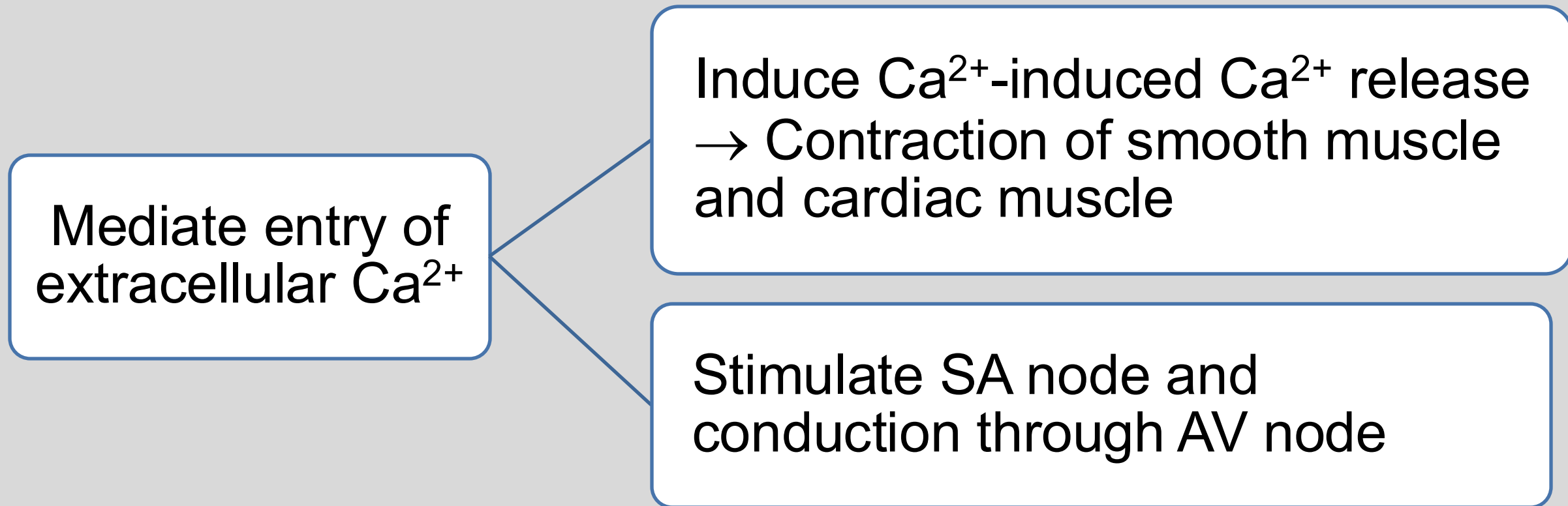
Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann:
 Goodman & Gilman's: The Pharmacological Basis of Therapeutics,
 Thirteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.

Graphs show plasma levels (C_p values) of amlodipine (left) and of nifedipine (right) in immediate-release (red) and slow-release (black) preparations; doses were administered at zero time.

The legend for this slide is in the notes section.

Actions L-type Voltage Gated Ca^{2+} Channels

- L-type Ca^{2+} channels (slow channels) have a long, large, high threshold
- Dominant type in vascular smooth muscle, cardiac myocytes, SA and AV nodal tissue



Influx of Ca^{2+} through L-type voltage-gated Ca^{2+} channels triggers smooth muscle contraction

calcium-calmodulin complex
activates MLCK



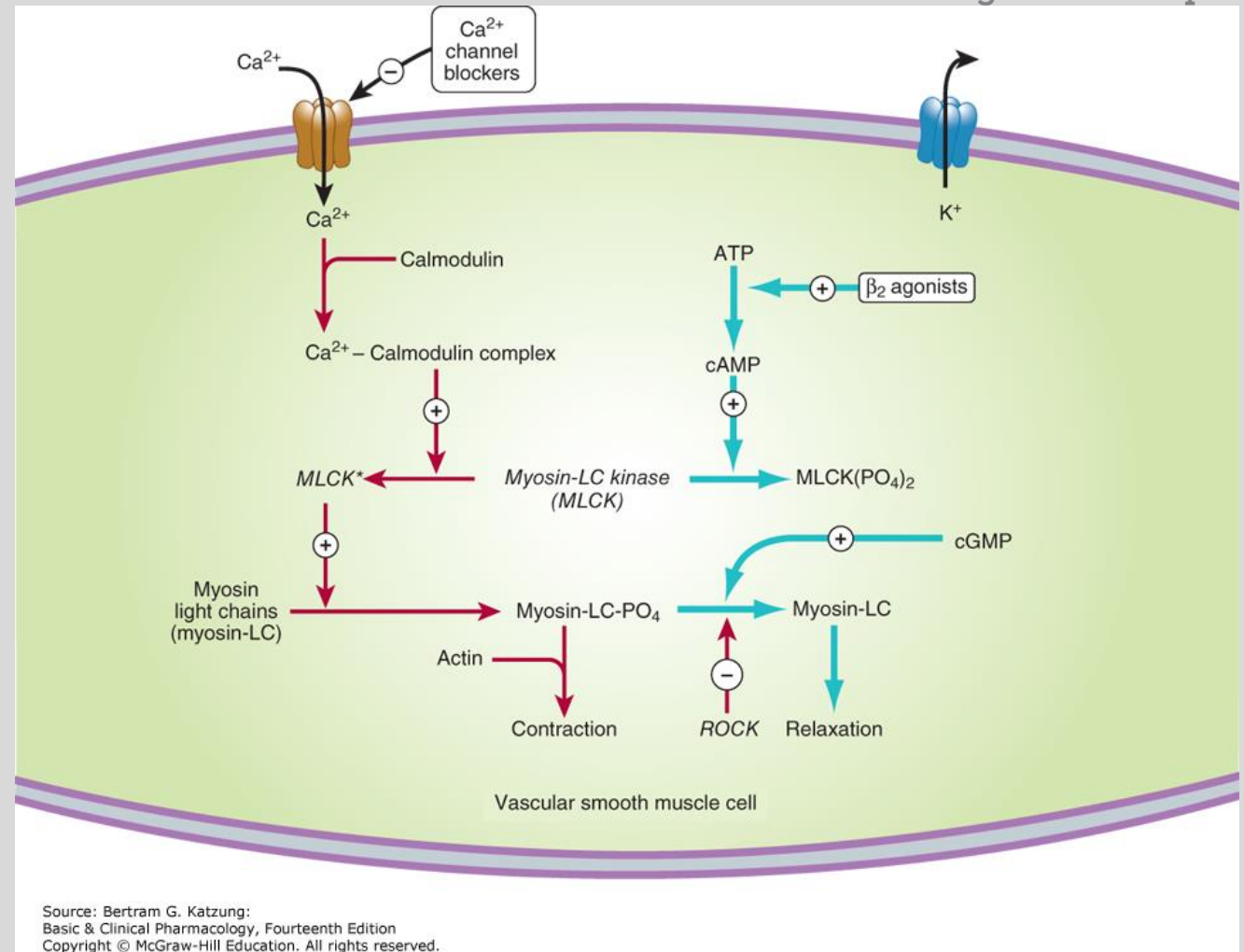
phosphorylation of myosin light
chains



myosin-actin interaction

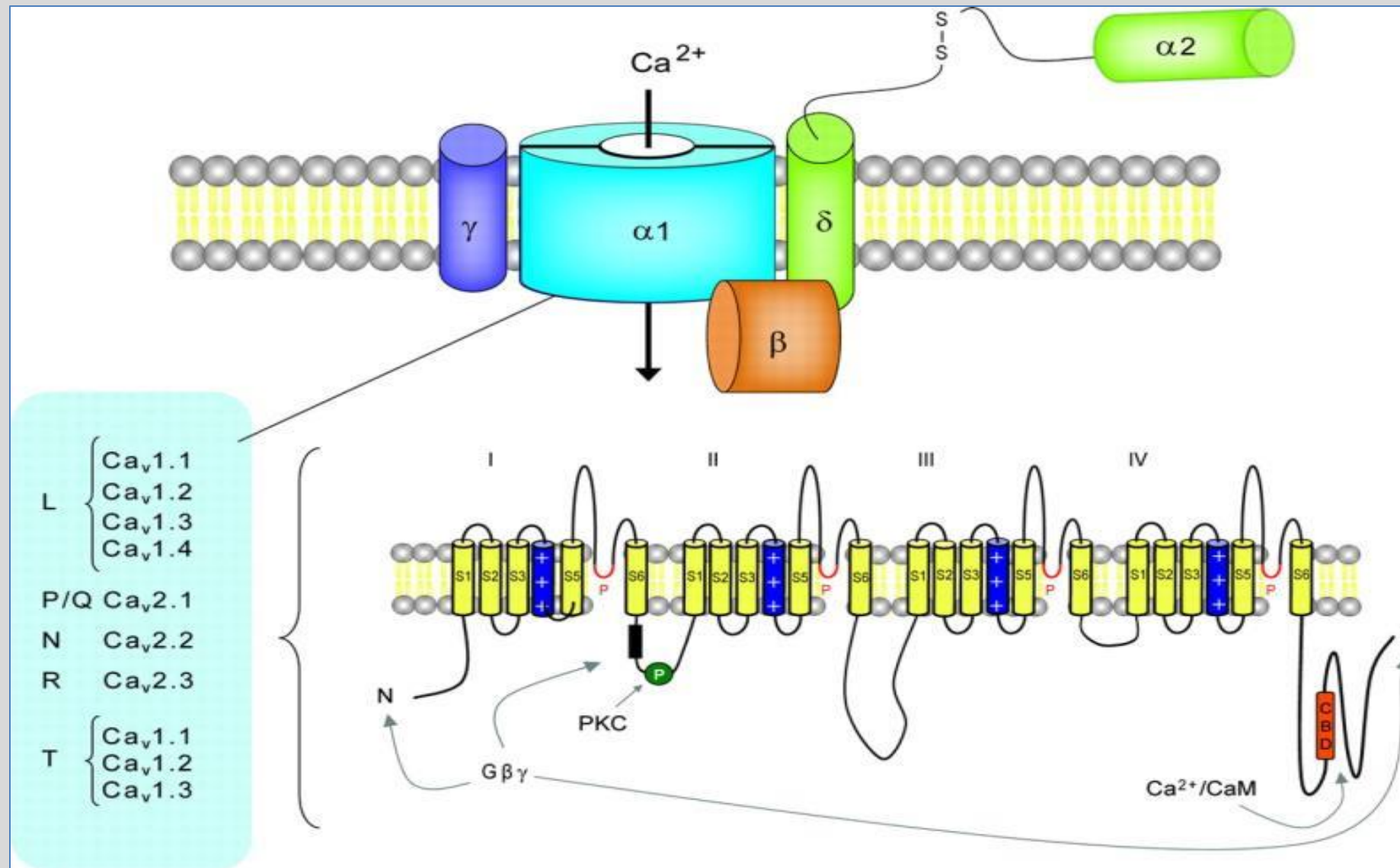


smooth muscle contraction



Calcium channel blockers inhibit this reaction leading to vasodilation

CCBs binding to L-type VG Ca^{2+} channel



- CCBs bind to a [^] site on the main pore-forming (α_1) subunit of the Ca^{2+} channel → block Ca^{2+} entry

The different CCBs bind different sites on the subunit

Access Medicine Katzung & Vanderah's Basic & Clinical Pharmacology, 15e, 2021; Chapter 13: Drugs Used in Heart Failure

Hess EJ, Jen JC, Jinnah HA, Benarroch EE. Neuronal voltage-gated calcium channels: brief overview of their functions and clinical implications in neurology

Neurology. 2010 Sep 7;75(10):937

CCBs act from the inner side of the membrane.
The drugs bind more effectively **to open channels and inactivated channels**.

Similar to Na⁺
channel block by
lidocaine

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Drug binding reduces the frequency of opening
in response to depolarization



marked decrease in
transmembrane calcium current

DHPs and non-DHPs

verapamil, diltiazem

Long-lasting arterial
smooth muscle relaxation

Reduction in
contractility
throughout the heart

Decreases in
SA node pacemaker
rate and
AV node conduction
velocity

CCBs: Dose-dependent effects

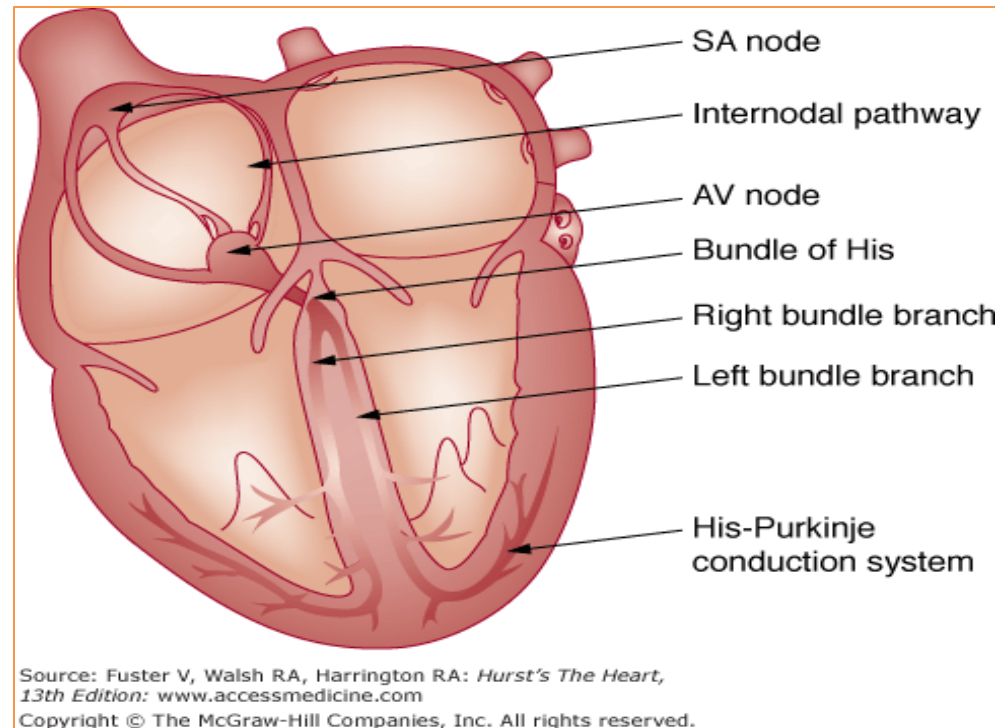
Verapamil / Diltiazem

- Decrease the channel's rate of recovery
- Channel block is enhanced as the frequency of stimulation increases
 - verapamil > diltiazem
- Depress rate of the sinus node pacemaker (automaticity)
- Slow AV conduction

Dihydropyridines (DHPs)

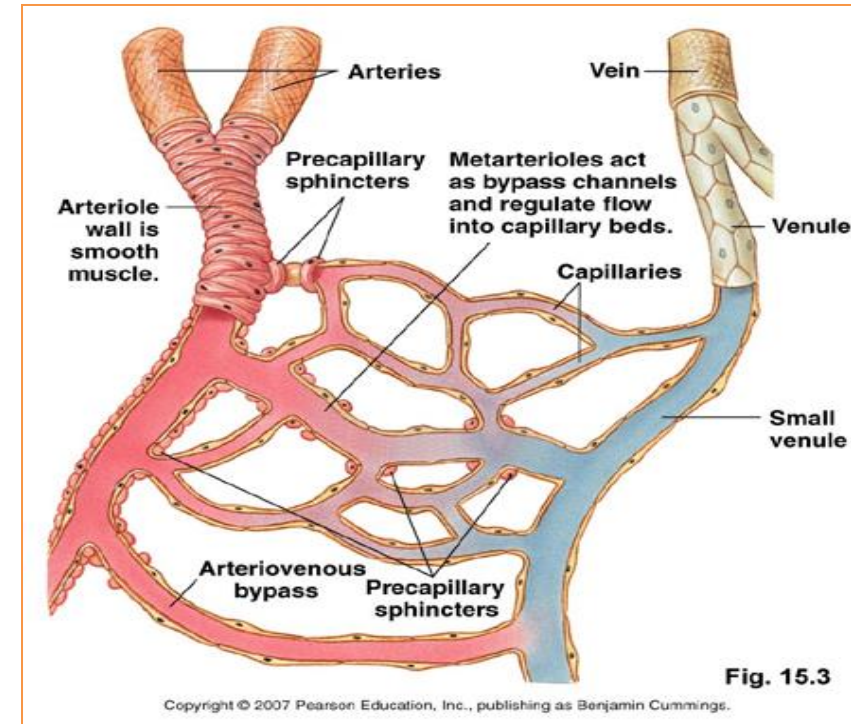
- Potent vasodilators
 - Reflex tachycardia and positive inotropy
 - More potent vasodilators than verapamil / diltiazem
- (DHPs > verapamil > diltiazem)
- Do not affect the channel's rate of recovery
- Do not affect conduction through the AV node

Calcium antagonists' cardiac and vascular effects



(verapamil and diltiazem)

- Slow heart rate
- Slow AV conduction
- Reduce myocardial contractility



(DHPs and non-DHPs)

↓ peripheral arterial resistance

↓ afterload

↓ O₂ consumption

Left : Access Medicine Hurst's The Heart, 13e, 2011; Figure 43-1 Schematic representation of the cardiac conduction system

Right: 2012 Pearson Education, Inc. Figure 21-5a The Organization of a Capillary Bed

<http://www.slideshare.net/gwrandall/ch-21lecturepresentation-25361230>

CCBs Therapeutic Uses and Effects

Hypertension
all CCBs
initial or add-on therapy

relax smooth muscle of arterioles



↓ systemic vascular resistance



↓ BP

Hypertensive emergency
nicardipine, clevidipine
(slow I.V. infusion)

- DHP CCB *plus* labetalol or esmolol to prevent reflex tachycardia
- Nicardipine: Potent, long-acting → duration ≤ 8 hours
- Clevidipine: 1-minute $t_{1/2}$ → duration 5-15 min

JNC8, black patients: CCB or thiazide diuretic recommended 1st-line
2025 ACC/AHA High Blood Pressure Guideline First-line therapy is the same for all patients.
Evidence supports individualized therapy based on comorbidities.

Angina

initial or add-on therapy

↓ peripheral vascular resistance
→ ↓ afterload



↑ coronary blood flow



↑ O₂ delivery to myocardium

Variant (vasospastic) angina:
CCBs induce coronary vasodilation.

Variant angina (coronary vasospasm)	• CCBs 1st-line • may combine with NTG
Exertional angina HR= heart rate SBP = systolic blood pressure	• BB <i>plus</i> DHP CCB more effective than either agent alone • Non-DHP act on the double product (HR x SBP): ↓HR x ↓SBP → ↓ O₂ demand
Unstable angina (acute coronary syndrome)	• DHP CCB only when combined with BB • Non-DHP only for patients who do not tolerate BB, do not have left ventricular dysfunction, and show no signs of disturbed AV conduction

CCBs Additional Therapeutic Uses and Effects

Supraventricular tachyarrhythmias	Verapamil and diltiazem (hemodynamically stable patients) to slow automaticity and slow conduction through AV node, which may slow ventricular response.
Raynaud phenomenon	Long-acting DHP CCBs. At least moderate reduction in intensity of attacks and prevention of tissue loss is achievable in most patients.
Subarachnoid hemorrhage	Nimodipine (lipophilic, cerebral artery vasodilation) to reduce incidence and severity of ischemic deficits
Migraine	Verapamil may prevent migraine (weak and conflicting data)
Tocolytic	Nifedipine (for 48 hours only) to suppress premature contractions in preterm labor (24-34 weeks)

CCBs Adverse Effects

non-DHP

Bradycardia and worsening cardiac output (usually with I.V.)

Constipation (especially verapamil)

DHP

Peripheral edema
(most common side effect)

Vasodilation → pre-capillary dilation and post-capillary reflex constriction → ↑ capillary pressure and permeability → redistribution of fluid into interstitium → edema

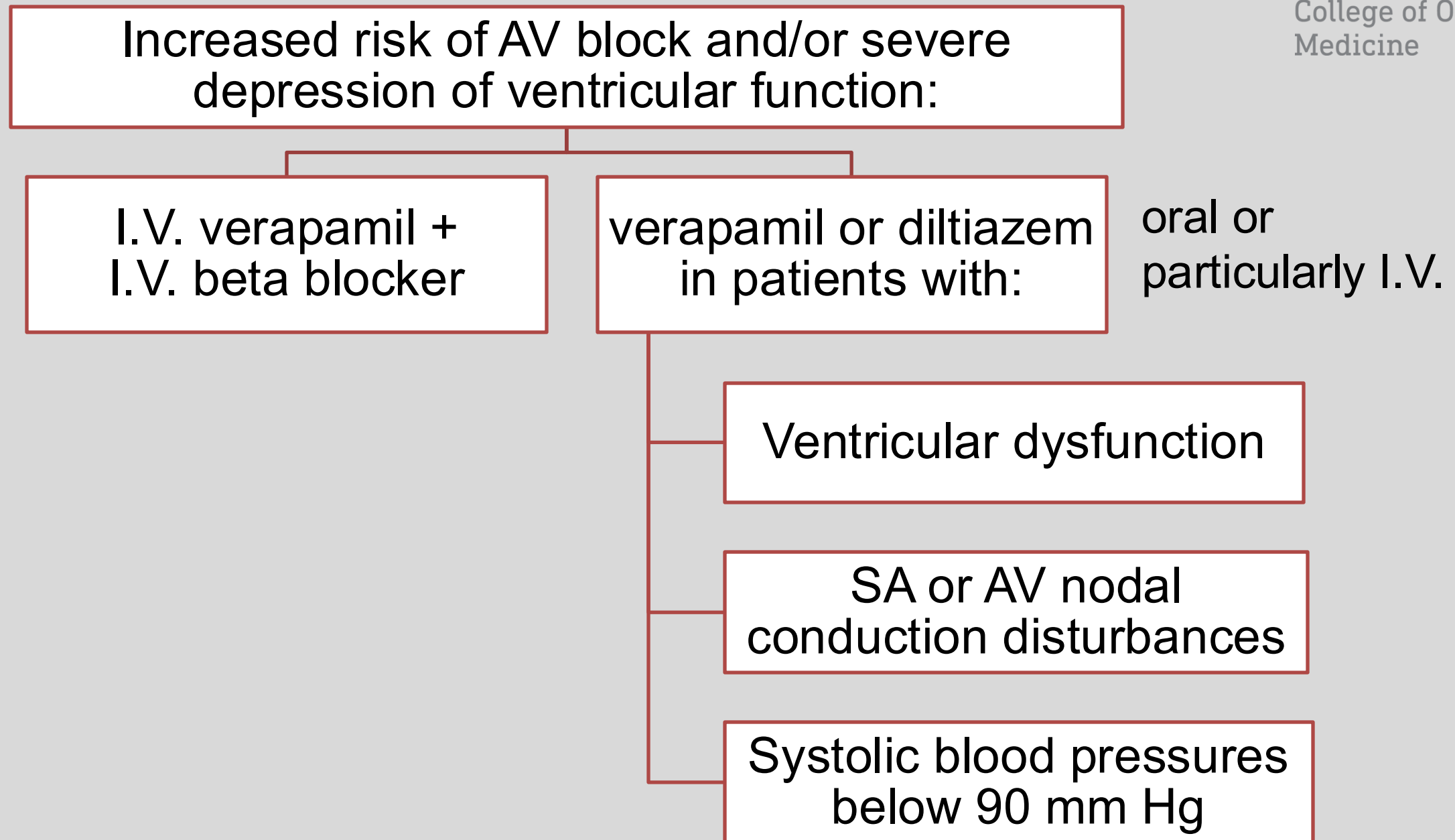
Both non-DHP and DHP:

GERD aggravation

Immediate-release:

- Headache, flushing, dizziness
- Hypotension – reflex tachycardia (sympathetic activation)

Verapamil / Diltiazem Contraindications



Nifedipine (immediate-release) Contraindications

Nifedipine immediate-release is not for use in patients with acute or unstable angina and STEMI

It may precipitate / aggravate myocardial infarction

- Causes abrupt vasodilation with reflex sympathetic activation
- MI may result from excessive hypotension, decreased coronary perfusion, or increased O₂ demand due to increased sympathetic tone and excessive tachycardia

Nifedipine short-acting formulations are not appropriate in the long-term treatment of angina or hypertension.

CCBs Drug Interactions

Verapamil + digoxin → digoxin toxicity

verapamil blocks Pgp → ↑ digoxin (Pgp substrate) levels → ↑ risk of digoxin toxicity

Pgp: P-glycoprotein

Verapamil + quinidine → excessive hypotension

- CYP3A4, Pgp inhibition, PD effects
- patients with idiopathic hypertrophic subaortic stenosis are at particular risk

CYP- and P-glycoprotein mediated interactions

Pharmacodynamic drug interactions

BETA BLOCKERS

(β adrenergic receptor antagonists)

* Propranolol * Metoprolol

* Labetalol * Esmolol

Carteolol (topical only)

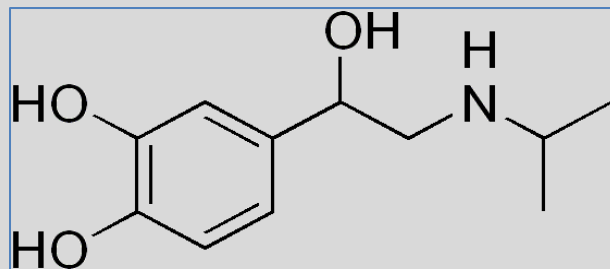
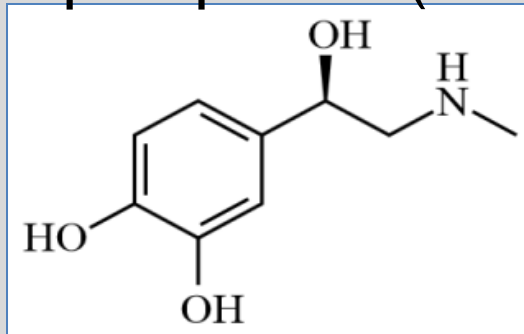
* Carvedilol

(and many more)

Beta Blockers: Structure-Activity Relationship

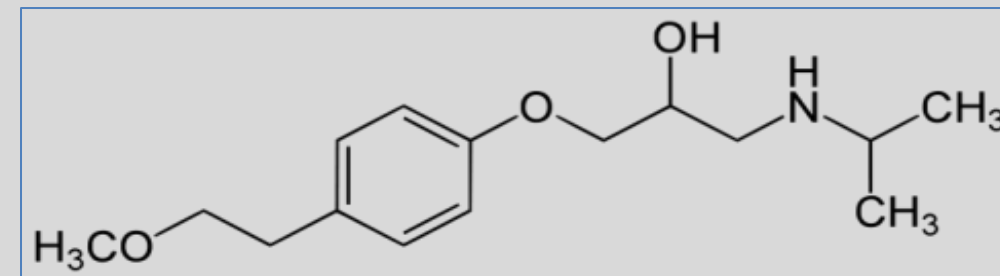
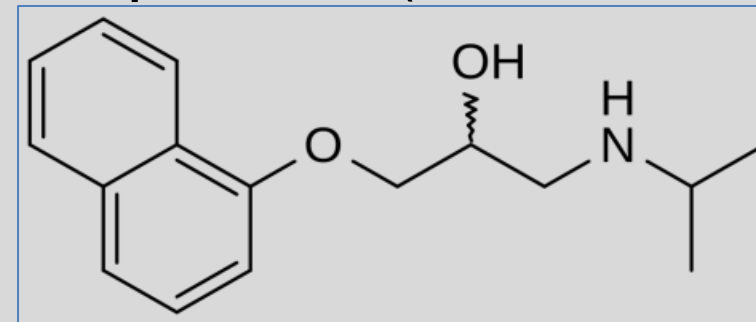
- Substitution of an isopropyl group or other bulky substituent on the amino nitrogen favors interaction with β receptors
- Most therapeutic beta blockers resemble isoproterenol to some degree

Epinephrine (catecholamine)



Isoproterenol (β AR agonist)

Propranolol (nonselective competitive antagonist)



Metoprolol (may be an inverse agonist)

Beta blockers are competitive antagonists of norepinephrine and epinephrine (or inverse agonists)

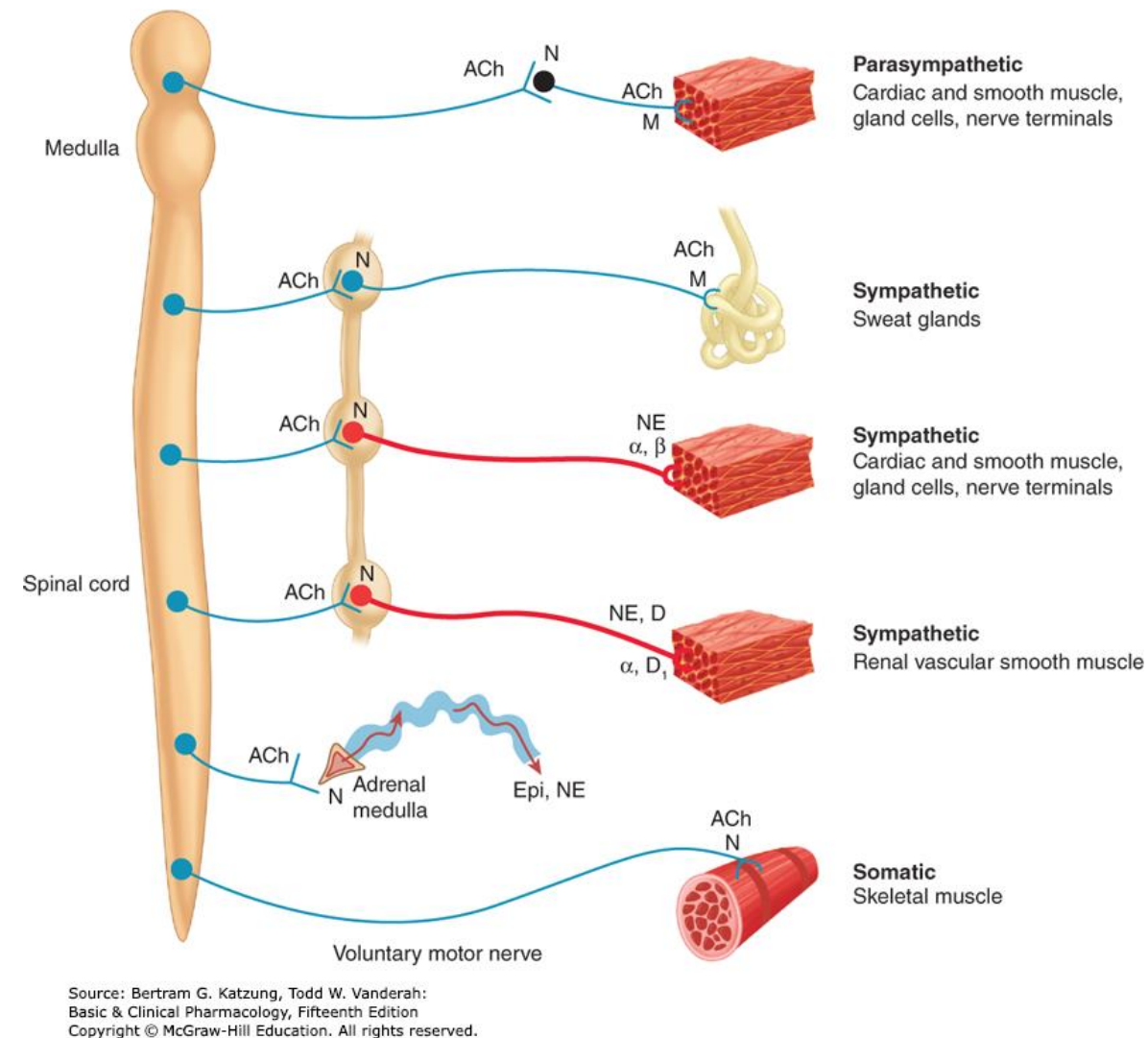
*Recognizing the
pharmacological
properties of beta
blockers through:*

Knowledge of the
responses elicited
by the receptors in
the various tissues

The activity of the
sympathetic
nerves that
innervate these
tissues

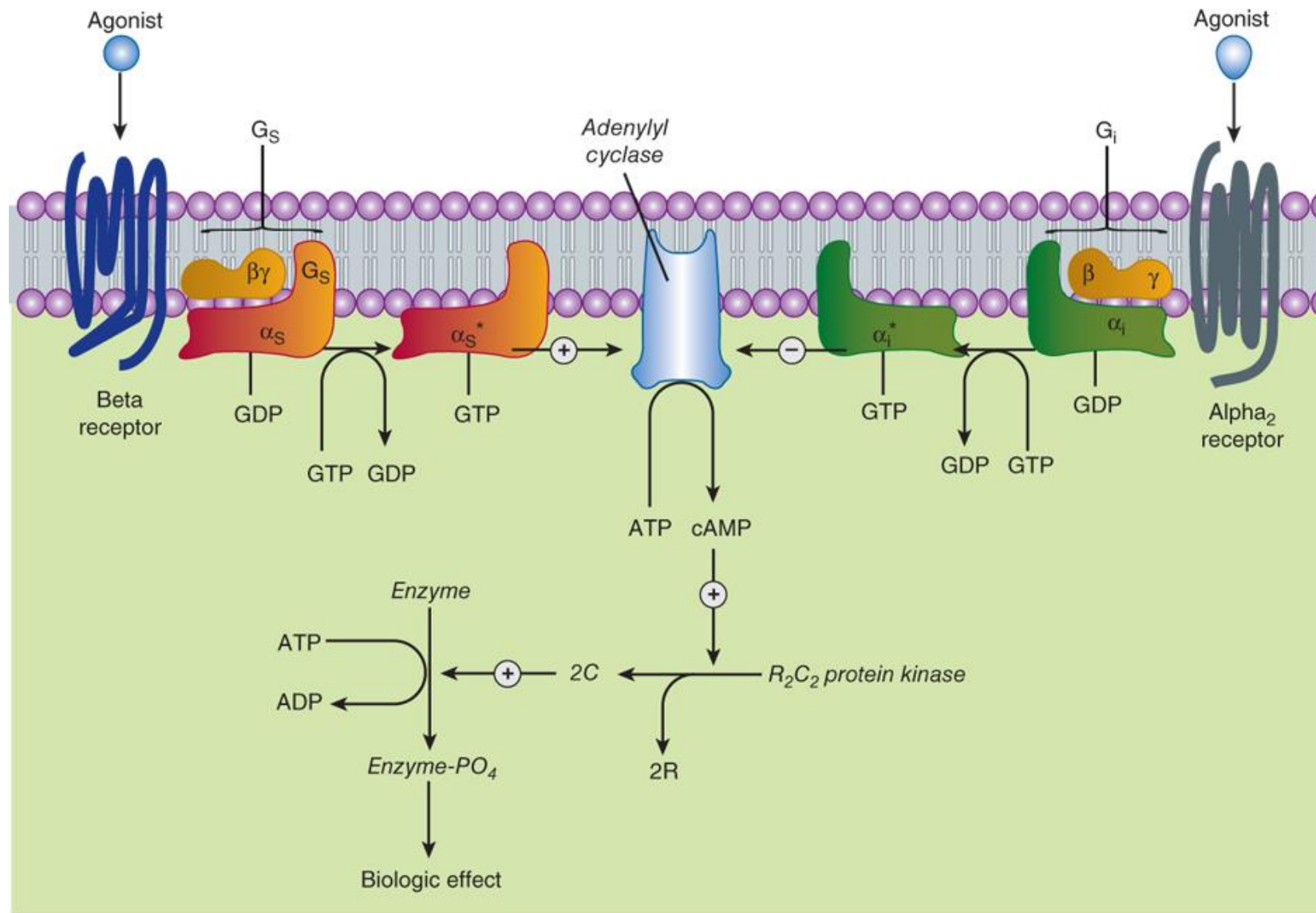
The
pharmacological
properties of β
receptor
antagonists

Some anatomic and neurotransmitter features of autonomic and somatic motor nerves



Schematic diagram comparing some anatomic and neurotransmitter features of autonomic and somatic motor nerves. Only the primary transmitter substances are shown. Parasympathetic ganglia are not shown because most are in or near the wall of the organ innervated. Cholinergic nerves are shown in blue, noradrenergic in red. Note that some sympathetic postganglionic fibers release acetylcholine rather than norepinephrine. Sympathetic nerves to the renal vasculature and kidney may release dopamine as well as norepinephrine during stress. The adrenal medulla, a modified sympathetic ganglion, receives sympathetic preganglionic fibers and releases epinephrine and norepinephrine into the blood. Not shown are the sacral preganglionic fibers that innervate the rectum, bladder, and genitalia. These fibers are probably sympathetic preganglionic nerves with cholinergic postganglionic fibers (see Box: Sympathetic Sacral Outflow). ACh, acetylcholine; D, dopamine; Epi, epinephrine; M, muscarinic receptors; N, nicotinic receptors; NE, norepinephrine.

β -Adrenergic receptors ($\beta_1\beta_2\beta_3$) are G_s PCRs: Agonist activation of the receptor activates adenylyl cyclase which increases cAMP, which activates PKA.



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
Copyright © McGraw-Hill Education. All rights reserved.

β ARs regulate numerous functional responses, including heart rate and contractility, smooth muscle relaxation and multiple metabolic events in numerous tissues.

Activation and inhibition of adenylyl cyclase by agonists that bind to catecholamine receptors. Binding to β adrenoceptors stimulates adenylyl cyclase by activating the stimulatory G protein, G_s , which leads to the dissociation of its α subunit charged with GTP. This activated α_s subunit directly activates adenylyl cyclase, resulting in an increased rate of synthesis of cAMP. Alpha₂-adrenoceptor ligands inhibit adenylyl cyclase by causing dissociation of the inhibitory G protein, G_i , into its subunits—ie, an activated α_i subunit charged with GTP and a β - γ unit. The mechanism by which these subunits inhibit adenylyl cyclase is uncertain. cAMP binds to the regulatory subunit (R) of cAMP-dependent protein kinase, leading to the liberation of active catalytic subunits (C) that phosphorylate specific protein substrates and modify their activity. These catalytic units also phosphorylate the cAMP response element binding protein (CREB), which modifies gene expression. See text for other actions of β and α_2 adrenoceptors.

Catecholamine Effects: Stimulating Cardiac Beta Receptors

Tissue	Effect	Receptors
Sinoatrial node	↑ heart rate ++	$\beta_1 > \beta_2$
Atria	↑ automaticity and conduction velocity ++	$\beta_1 > \beta_2$
Atrioventricular node	↑ automaticity and conduction velocity ++	$\beta_1 > \beta_2$
His-Purkinje system	↑ automaticity and conduction velocity	$\beta_1 > \beta_2$
Ventricles	↑ contractility, conduction velocity, automaticity and rate of idioventricular pacemakers +++	$\beta_1 > \beta_2$

Note: Parasympathetic effect on the SA node, atria, and AV node (via stimulation of cardiac M₂ receptors) is the opposite of the adrenergic effect; parasympathetic stimulation has only slight effect on the His-Purkinje system and ventricle.

Responses + to +++: Approximate indication of the importance of the sympathetic activity in controlling the functions listed

Physiologic Effects of Stimulating Beta Receptors in:

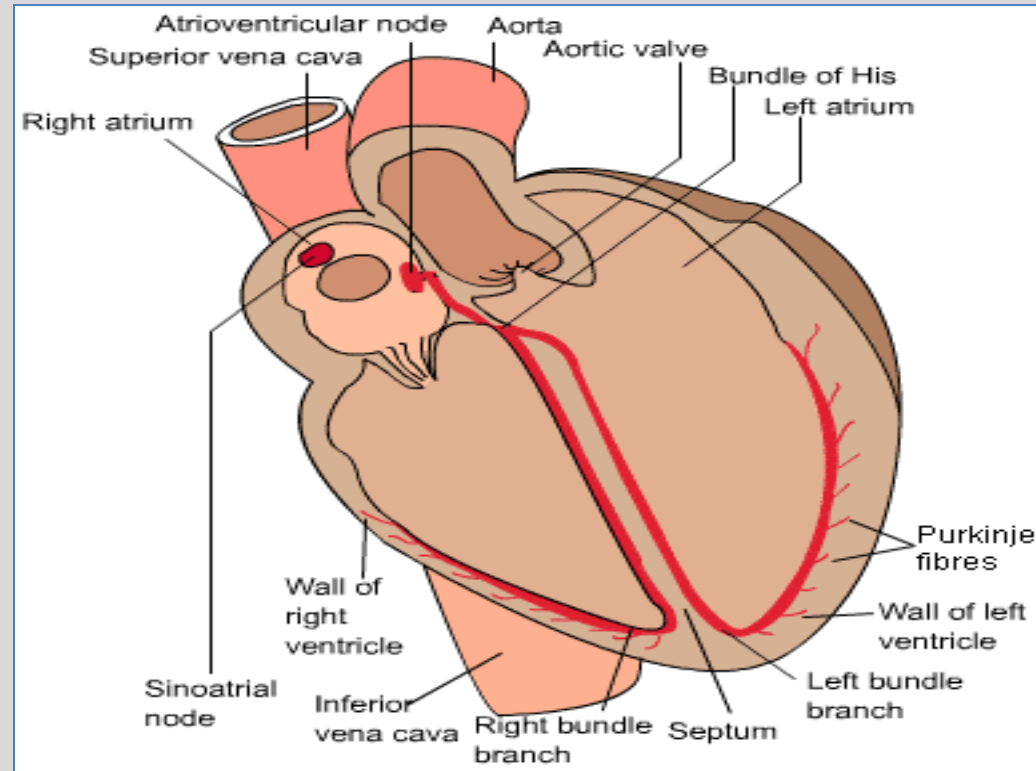
- Skeletal muscle and hepatic vasculature:

The β_2 receptor response — vasodilation — predominates

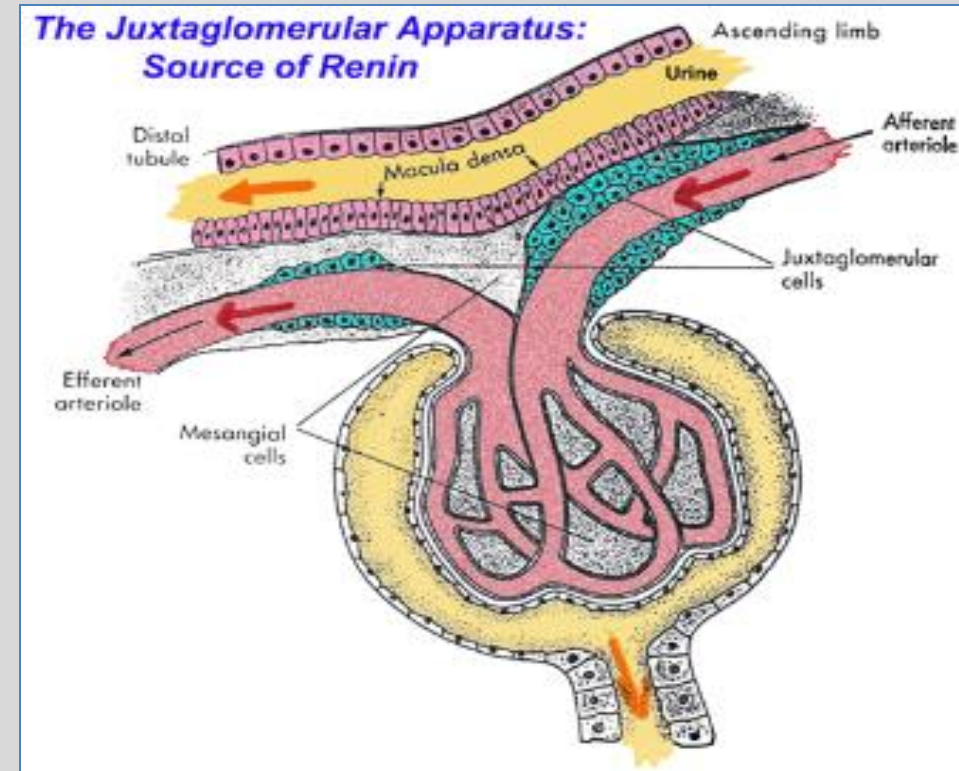
- Renal juxtaglomerular (JG) apparatus

β_1 receptor activation induces renin release → activation of the renin-angiotensin-aldosterone system

Beta blockers reduce blood pressure in hypertensive individuals by reducing cardiac output and renin secretion:



hsc.uwe.ac.uk/rcp/cs-heart-inside.aspx



www.marcoantonioramos.com/physiology1/reninse...

- Blocking cardiac β_1 adrenergic receptors $\rightarrow$ slows heart rate, AV nodal conduction, and myocardial contractility $\rightarrow \downarrow$ CO
- Blocking renal β_1 adrenergic receptors $\rightarrow$ inhibits renin release

Classical non-selective beta blockers: First generation

DRUG	MEMBRANE STABILIZING ACTIVITY	INTRINSIC AGONIST ACTIVITY	LIPID SOLUBILITY	EXTENT OF ABSORPTION (%)	ORAL AVAILABILITY (%)	PLASMA t _{1/2} (hours)	PROTEIN BINDING (%)
Nadolol	0	0	Low	30	30-50	20-24	30
Penbutolol	0	+	High	~100	~100	~5	80-98
Pindolol	+	+++	Low	>95	~100	3-4	40
Propranolol	++	0	High	<90	30	3-5	90
Timolol	0	0	Low to moderate	90	75	4	<10

β₁-Selective beta blockers: Second generation

DRUG	MEMBRANE STABILIZING ACTIVITY	INTRINSIC AGONIST ACTIVITY	LIPID SOLUBILITY	EXTENT OF ABSORPTION (%)	ORAL AVAILABILITY (%)	PLASMA t _{1/2}	PROTEIN BINDING (%)
Acebutolol	+	+	Low	90	20-60	3-4 h	26
Atenolol	0	0	Low	90	50-60	6-7 h	6-16
Bisoprolol	0	0	Low	≤90	80	9-12h	~30
Esmolol	0	0	Low	NA	NA	9 min	55
Metoprolol	+ *	0	Moderate	~100	40-50	3-7 h	12

*Detectable only at doses much greater than required for β blockade
h: hours

Beta blockers with additional actions:

Third generation

DRUG	MEMBRANE STABILIZING ACTIVITY	INTRINSIC AGONIST ACTIVITY	LIPID SOLUBILITY	EXTENT OF ABSORPTION (%)	ORAL AVAILABILITY (%)	PLASMA t _{1/2} (hours)	PROTEIN BINDING (%)
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Non-selective β blockers with additional actions:

Carteolol	0	++	Low	85	85	6	23-30
Carvedilol	++	0	Moderate	>90	~30	7-10	98
Labetalol	+	+	Low	>90	~33	~5	~50

β_1 -selective β blockers with additional actions:

Betaxolol	+	0	Moderate	>90	~80	15	50
Celiprolol	0	+	Low	~74	30-70	5	4-5
Nebivolol	0	0	Low	NA	NA	11-30	

Third generation beta blockers

with putative additional mechanisms of vasodilation

NITRIC OXIDE PRODUCTION	β_2 RECEPTOR AGONISM	α_1 RECEPTOR ANTAGONISM	Ca ²⁺ ENTRY BLOCKADE	K ⁺ CHANNEL OPENING	ANTIOXIDANT ACTIVITY
Nebivolol		Carvedilol	Carvedilol		Carvedilol
Carteolol	Carteolol	Labetalol	Betaxolol		

If you treat patients visiting from other countries, you may see the following BBs, which are not currently available in the U.S.:

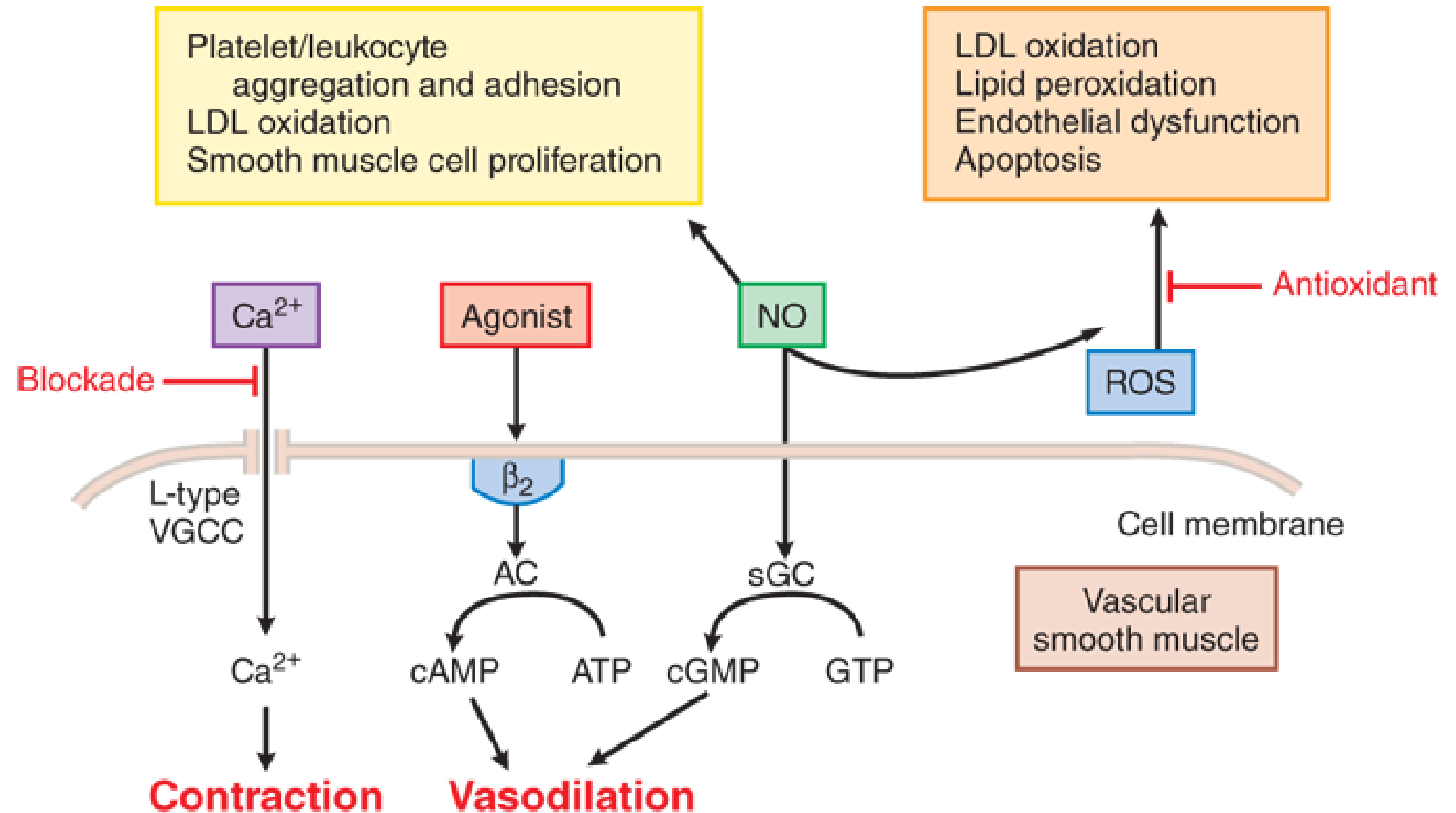
Celiprolol	Celiprolol	Bucindolol	Bevantolol	Tilisolol
Bopindolol	Bopindolol	Bevantolol		
Nipradilol		Nipradilol		

Mechanisms underlying actions of vasodilating β blockers in blood vessels.

AC: adenylyl cyclase;
sGC: soluble guanylyl cyclase;
NO: nitric oxide;
ROS: reactive oxygen species;
VGCC: voltage-gated Ca^{2+} channel.

(Modified with permission from Toda N.
Vasodilating β adrenoceptor blockers as
cardiovascular therapeutics.

Pharmacol Ther, 2003, 100:215–234.
Copyright © Elsevier.)



Source: Laurence L. Brunton, Björn C. Knollmann:
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:
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Beta blockers are recommended for hypertensive patients with concomitant heart disease and related comorbidities:

- Hypertension
- Hypertensive emergencies: labetalol, esmolol (other classes of vasodilators are also used)
- Intraoperative tachycardia, hypertension: eg propranolol, metoprolol, esmolol
- Ischemic heart disease (angina)
- Congestive heart failure
- Certain arrhythmias
- Hyperthyroidism
- Glaucoma (topical ophthalmic drops)

Labetalol is a recommended agent for the treatment of hypertensive emergency in pregnant persons.



Mechanistic rationales for the use of beta blockers in each of the disorders.

Hypertension: Beta Blockers Mechanism of Action

Hypertension is initiated and sustained at least in part by sympathetic neural activation.



Drugs, such as beta blockers, which block catecholamine actions, have important clinical effects.

- BBs reduce cardiac output and lower blood pressure in patients with hypertension (when taken regularly)
- Mechanisms probably include suppression of renin release and effects in the CNS

Hypertension: Beta Blockers Effects

- β receptor blockade has relatively little effect on the *normal heart* of an individual at rest...
- ...But, beta block has profound effects when sympathetic control of the heart is dominant, as during exercise or stress.
- BBs do *not usually* cause hypotension in healthy individuals with normal blood pressure.

Ischemic Heart Disease: Beta Blockers Mechanism of Action

negative chronotropic effect
(particularly during exercise)

+

negative inotropic effect

+

reduction in
arterial blood pressure
(particularly systolic pressure)
during exercise

reduce myocardial O₂
consumption

at rest and during exertion

Ischemic Heart Disease: Beta Blockers Effects

In the treatment of exertional angina:

- BBs reduce myocardial O₂ consumption at rest and during exertion
 - also, they cause some increase in flow toward ischemic regions by increasing collateral circulation
- BBs have shown to:
 - Reduce the severity and frequency of attacks of exertional angina, and
 - Improve survival in patients who have had an MI

Ischemic Heart Disease: Beta Blockers Cautions

BBs' undesirable effects in angina include:

- ↑End-diastolic volume and
- ↑Ejection time

tend to increase
myocardial O₂
requirement

Nitrates counterbalance these deleterious effects of BBs.

Congestive Heart Failure: BBs Mechanism of Action

Suggested mechanisms include:

1. Decrease unstable tachyarrhythmias
2. Attenuate sustained sympathetic activation
 - inhibit maladaptive proliferative cell signaling in the myocardium
 - reduce catecholamine-induced cardiomyocyte toxicity
 - decrease myocyte apoptosis
3. Reduced remodeling
 - decrease LV chamber size and increase LV ejection fraction
 - may decrease oxidative stress in the myocardium

Congestive Heart Failure: BBs Effects

Most patients with chronic heart failure respond favorably to certain β blockers.

- Caution: BBs can precipitate acute decompensation of cardiac function.

Only drugs shown to be effective in stable heart failure should be used.

- Studies with **bisoprolol**, **carvedilol**, **metoprolol** and **nebivolol** showed a reduction in mortality in patients with stable severe heart failure.
- (This effect was not seen with bucindolol.)

Arrhythmias: Beta Blockers Mechanism of Action

1. Reduce heart rate
2. Decrease intracellular Ca^{2+} overload
3. Inhibit afterdepolarization-mediated automaticity
4. Increase AV nodal conduction time (increased PR interval)
5. Prolong AV nodal refractoriness
 - Some exert Na^+ channel–blocking effects (membrane stabilizing effects) at high concentrations
 - Examples: propranolol, carvedilol
 - BBs are useful in terminating re-entrant arrhythmias involving the AV node and in controlling ventricular response in atrial fibrillation or flutter.
 - BBs can prevent recurrent infarction and sudden death in patients recovering from acute myocardial infarction.

Beta blockers are
Vaughan-Williams
class II
antiarrhythmics

Hyperthyroidism: Beta Blockers Effects

Hyperthyroidism increases the basal metabolic rate and enhanced organ sensitivity to catecholamines

- It is associated with an increased number of β -adrenergic receptors.

BBs ameliorate symptoms:

- e.g. palpitations, tachycardia, tremulousness, anxiety, heat intolerance

Thyroid storm can cause supraventricular tachycardia and sudden death.

- clinical manifestations of thyrotoxicosis

BBs control:

- pulse rate
- hypertension
- atrial fibrillation

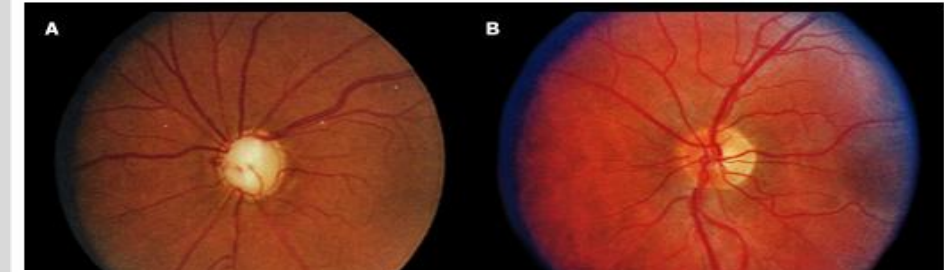
Glaucoma: Ophthalmic Beta Blockers

Mechanism of Action and Effects

- β receptors of the eye largely of the β_2 subtype
 - Located in the ciliary body epithelium and blood vessels
- BBs decrease IOP presumably by:
 - ↓ Production of aqueous humor, and/or
 - ↓ Ocular blood flow, which decreases the ultrafiltration responsible for aqueous production

IOP: intraocular pressure

Glaucoma



- A) Extensive cupping of the optic disk typical of glaucoma
- B) Compared to a normal optic disk

Courtesy of the W.K. Kellogg Eye Center, University of Michigan.

UpToDate®

- Little or no effect on pupil size or accommodation
- Do not cause blurred vision or night blindness
- Nasolacrimal drainage $\Rightarrow$ **BBs potentially can cause adverse cardiovascular and pulmonary effects in susceptible patients.**

Situational / Performance Anxiety: Beta Blockers

- Situational or performance anxiety – “stage fright”: Extreme anxiety or panic attacks in specific situations
- Sympathetic manifestations may include palpitations, tightness in the chest, breathlessness, muscular tension, trembling, flushing, sweating, dry mouth
- Beta blockers:

Low dose of propranolol taken an hour before a performance reliably decreases somatic manifestations of anxiety, such as the pounding heart and trembling hands and voice.

Beta Blockers Adverse Effects and Mechanisms

Excessive bradycardia	Block of cardiac β receptors (risk of bradyarrhythmias in patients with AV conduction defects)
Peripheral vascular insufficiency	Nonselective agents block β 2-mediated peripheral vasodilation. (β 1-selective agents are better tolerated in mild-moderate disease. But, caution in severe peripheral vascular disease or vasospastic disorders.)
Asthma / COPD exacerbation	Block β 2-mediated bronchodilation → Airway resistance / Bronchospasm
Delayed recovery from insulin-induced hypoglycemia	1) Inhibit β 2-mediated hepatic glycogenolysis / gluconeogenesis 2) Blunt perception of symptoms (tremor, tachycardia, nervousness) • β 1-selective agents may be less prone to inhibit recovery from hypoglycemia

Beta Blockers Adverse Effects and Mechanisms

Decrease insulin sensitivity (non-vasodilating agents)	Mechanisms not understood • Can occur with both selective and nonselective agents
Increase plasma VLDL and decrease HDL	
Decrease release of free fatty acids from adipose tissue	Block β -mediated activation of hormone-sensitive lipase in fat cells (free fatty acids are an energy source for exercising muscle)
CNS effects	Fatigue; insomnia; nightmares (related to the drug's lipophilicity? No clear correlation has emerged.)
Decreased libido	Mechanism not understood

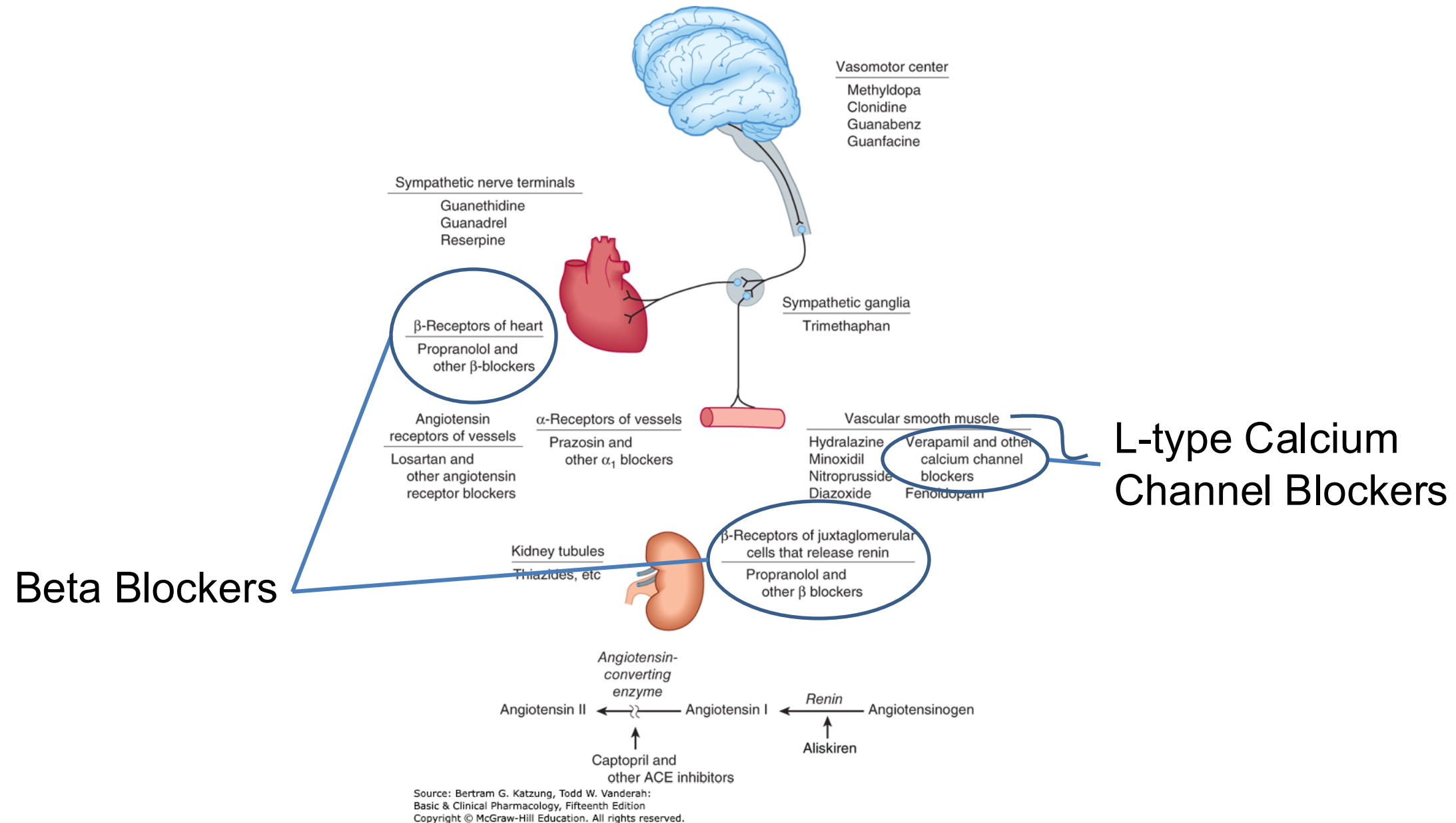
Beta Blockers Withdrawal Syndrome

- Nervousness
- Tachycardia
- Increased blood pressure
- Increased intensity of angina
- Increased risk of sudden death
MI reported in a few patients
- The withdrawal syndrome may involve upregulation or supersensitivity of β adrenoceptors
- **BBs should not be abruptly discontinued after prolonged regular use.**

Summary of calcium channel blockers and beta blockers pharmacology

Sites of action of the major classes of antihypertensive drugs.

Most antihypertensive agents work by reducing cardiac output and/or decreasing peripheral resistance.



- Calcium channel blockers (CCBs) lower blood pressure by relaxing arteriolar smooth muscle and decreasing peripheral vascular resistance. Non-DHP CCBs Verapamil and diltiazem also decrease cardiac rate and force and reduce oxygen demand.
- CCBs are recommended first-line agents in the treatment of primary hypertension.
- Beta blockers lower blood pressure by decreasing cardiac output and cardiac contractility, and they reduce oxygen demand.
- Diuretics, ACEI, ARBs, CCBs, BBs, and alpha blockers are essentially equivalent in blood pressure lowering effect.
- Antihypertensives are frequently used in combination for increased efficacy.
- CCBs and beta blockers are used in the treatment of angina, supraventricular arrhythmias, and other uses.
- Side effects are related to the mechanisms.

Summary: Calcium Channel Blockers

Non-DHP CCBs: Verapamil, Diltiazem

- Oral, IV, duration 4 to 24h (long-acting)
- Block L-type calcium channels in vessels and heart
- Reduce vascular resistance, cardiac rate, and cardiac force, which results in ↓ O₂ demand
- Used for angina prophylaxis, hypertension, rate control in atrial fibrillation, SVT treatment and prophylaxis
- Toxicity: AV block, acute heart failure; constipation; edema
- Interactions: Additive with other cardiac depressants and hypotensive drugs

DHP CCBs

- Oral, IV duration 4 to >24 h
- Block vascular L-type calcium channels to a greater extent than cardiac channels
- Reduce vascular resistance
less cardiac depressant effect than non-DHP CCBs
- Prophylaxis of angina and treatment of hypertension
Immediate release nifedipine is contraindicated.
- Toxicity: Excessive hypotension, baroreceptor reflex tachycardia
- Interactions: Additive with other vasodilators

Summary of Beta Blockers

- Oral and parenteral; 4 to 24h (extended release)
- Beta blockers competitively block beta-adrenergic receptors. Some nonselectively block beta-1 and beta-2 receptors; others are relatively selective for beta-1 receptors. Some have additional properties, such as partial agonist effects, membrane stabilizing activity (high-dose), alpha-1 receptor block, and other effects.
- They decrease heart rate, cardiac output, myocardial O₂ demand.
- They are used for prophylaxis of angina, post-MI, compensated heart failure, as adjunctive agents in the treatment of hypertension, and other uses.
- Toxicity: Asthma, atrioventricular block, acute heart failure, sedation; beta-1selective blockers have less, but still significant, risk of bronchospasm
- Interactions: Additive with all cardiac depressants

- Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024
 - Chapter 8: Cholinoceptor Blocking Drugs
 - Chapter 9: Adrenoceptor Agonists & Sympathomimetic Drugs
 - Chapter 10: Adrenoceptor Antagonist Drugs Chapter 11: Antihypertensive Agents
 - Chapter 13: Drugs Used in Heart Failure
- Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e, 2023
 - Chapter 10: Neurotransmission: The Autonomic and Somatic Motor Nervous Systems
 - Chapter 14. Adrenergic Agonists and Antagonists
 - Chapter 31: Treatment of Ischemic Heart Disease
 - Chapter 34. Anti-Arrhythmic Drugs
 - Chapter 74: Ocular Pharmacology
- Access Medicine Williams Obstetrics: 25e, 2018; Chapter 40
- Cochrane Review published Sept 11, 2013 (http://www.cochrane.org/CD005508/PVD_beta-blockers-for-peripheral-arterial-disease) (Slide 52)
- JNC8 Guidelines: <https://jamanetwork.com/journals/jama/fullarticle/1791497>
- Lexidrugs monographs: individual agents
- UpToDate: various articles; literature current through May 2024; references are listed in the articles

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